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Original research article

Research on the feedback control characteristics and parameter optimization of closed-loop fiber optic gyroscope

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A R T I C L E I N F O

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Closed-loop control model
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A B S T R A C T

The feedback parameter of closed-loop fiber optic gyroscope (FOG) in various engineering applications is discussed. It has been proved that unreasonable feedback parameter will cause great interference and serious distortion to the output signal. Therefore, we build a concise feedback control mode and obtain optimal range of feedback parameter. For different input signals concerned which nearly include all of the application scenarios in practice, detailed simulations and experiments are completed. The results show that the feedback parameter should be small in static applications and moderate for high response requirements. What is more, it should be relatively small when used in low-bandwidth measurement fields to ensure the response performance of low frequency signal and suppress high frequency noise and vice versa. This work plays an important guiding role for related parameter setting in the feedback control system of sensors, especially for gyroscope.

1. Introduction

Fiber optic gyroscope is a kind of all solid-state inertial instrument which is based on Sagnac effect [1] to measure rotation rate. Compared with traditional electromechanical gyroscope, FOG has the advantages of small size, low cost, long service life, large dynamic range and short start-up time and has been widely used in many civilian and military applications like aerospace attitude control, navigation orientation, land navigation, resource exploration and mining [2–7]. At present, with the help of advances in optic and electronic techniques, FOG can achieve the accuracy coverage from 100 deg/h to 0.001 deg/h while cost less than its main competitor, ring laser gyroscope (RLG) [8–12]. FOG can be divided into open-loop and closed-loop schemes according to the way of signal processing [13]. Because it is superior to open-loop FOG in scale factor nonlinearity, dynamic range and precision, all-digital closed-loop system with digital demodulation and step-wave feedback is usually used in medium and high precision FOG.

Closed-loop system is an automatic control mode composed of signal positive path and feedback channel. The measurement results are fed back to the input so that the difference between them is obtained to reduce measurement error. At the same time, the system forms a closed loop which effectively improves its accuracy and anti-interference ability. In the closed-loop FOG, rotation rate measurement value is added to the phase modulator in the form of feedback phase for good linear response. In recent years, a lot of

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detailed works have been done to improve the performance of closed-loop FOG. Chong published an analysis of dead zone sources in which any rotation cannot be detected for the closed-loop FOG [14]. Bacurau proposed and experimentally demonstrated a new technique for modulation [15]. Refs [16,17]. also studied modulation approach but in depolarized FOG or three-axis FOG. Çelikel established all digital closed-loop FOG prototype covering design details [18]. However, as far as we know, the research on feedback parameter has not formed a theoretical system which is often set by tedious test or engineering experience for most cases. It has been proved by this paper and engineering experience that unreasonable parameter will greatly affect the performance of FOG. To achieve its better application in different fields, the feedback control model and related parameter setting should be studied and designed more carefully.

In this paper, the digital signal control model of closed-loop FOG is established and different feedback parameters are simulated with three typical kinds of input signals which include most of our application scenarios: random signal, step signal and sinusoidal signal. The influence of feedback parameters on system is analyzed by comparing the characteristic of output signal with input signal. What is more, relevant experiments are completed to verify our theory and simulation.

2. Signal feedback control principal and model of closed-loop FOG

The structure of all-digital closed-loop FOG is shown in Fig. 1. The light source, photodetector, coupler, integrated optical phase modulator and fiber coil constitute the optical system, of which the function is to form optical interference information and sense rotation rate. After completing the sensing process of external physical quantity, the prefilter circuit will shape and amplify the signal and the analog signal is converted to the corresponding digital signal by analog-to-digital converter (ADC). The digital signal output will be carried out after demodulating the corresponding rotational speed information by the digital signal processor (DSP). On the other hand, the closed-loop feedback control signal is generated by DSP which is added to the Y waveguide after digital-to-analog convert (DAC) to realize the whole closed-loop feedback process.

The interference light intensity on the photodetector can be expressed as:

$$I = I_0(1 + \cos\varphi_n) \tag{1}$$

Here, I_0 is the intensity of positive and negative propagation light, φ_n is the nonreciprocal phase shift. Then its detection sensitivity is:

$$\frac{dI}{d\varphi} = I_0 \sin\varphi_n \tag{2}$$

For open-loop systems, φ_n is exactly equal to Sagnac phase shift φ_s. But in the closed-loop control system, φ_n is the sum of φ_s and the feedback phase shift φ_f. Because $\varphi_f \approx -\varphi_s$ should be satisfied for a stable closed-loop system, we define $\varphi_s + \varphi_f$ as the residual phase difference φ_r. To avoid detection sensitivity of the working phase shift area approaching 0 when $\varphi_n \approx 0$, a bias modulation phase φ_b is added to φ_n in general engineering and φ_n is expressed as:

$$\varphi_n = \varphi_r + \varphi_b \tag{3}$$

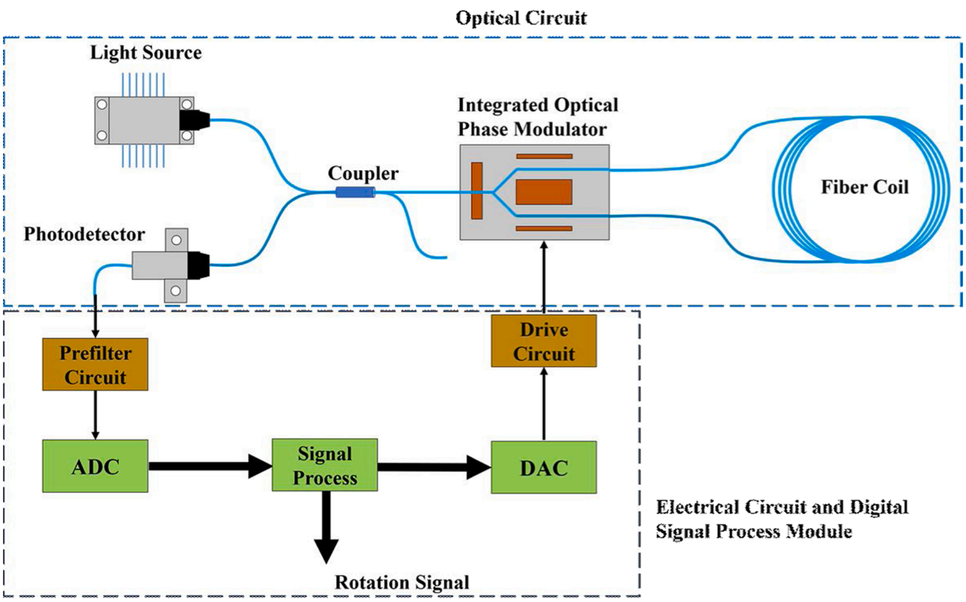


Fig. 1. Configuration of all-digital closed-loop FOG.

For two-state modulation, the principle of phase bias and signal demodulation is shown in Fig. 2. Apply the bias phase of φ_{b+} and φ_{b-} in two adjacent periods respectively where $\varphi_{b+} + \varphi_{b-} = 0$, and $I_+ = I_-$ can be satisfied when $\varphi_r = 0$. That is, if the intensity of light detected in two adjacent modulation periods is equal, the demodulated residual phase difference φ_r is 0. On the other hand, when $\varphi_r \neq 0$, its value can also be demodulated by the light intensity difference and the integral correction of detection value will be carried out. The above-mentioned light intensity difference can be expressed as:

$$\Delta I = I_+ - I_- = -2I_0 \sin\varphi_b \sin\varphi_r = k_0 \sin\varphi_r \tag{4}$$

For a stable closed-loop system, Eq. (4) can be turned into:

$$\Delta I = k_0\varphi_r = k_0(\varphi_s + \varphi_b) \tag{5}$$

Since the modulation of Y waveguide has an effect of time-delay difference, it is necessary to integrate the output digital quantity again before applying the modulation phase to obtain the corresponding rate ramp signal. The original signals of bias modulation, feedback modulation and rate ramp are as shown in the Fig. 3.

By abstracting the process of signal transmission and processing in closed-loop FOG, we get a simplified mathematical model as shown in Fig. 4.

Here k_1 includes the gain of forward channel including k_0, ADC factor, preamplifier, etc. k_2 includes the gain of feedback channel including DAC factor, electro-optic coefficient of Y waveguide, feedback parameter, etc. $I_1(Z)$ and $I_2(Z)$ are integrators and $D(Z)$ is differentiator. The transfer function of closed-loop control system is:

$$H(Z) = \frac{k_1 I_1(Z)}{1 + k_1 k_2 I_1(Z) I_2(Z) D(Z) Z^{-1}} \tag{6}$$

Since $I(Z) = \frac{1}{1-Z^{-1}}, D(Z) = 1 - Z^{-1}$, Eq. (6) can be simplified as:

$$H(Z) = \frac{k_1 I_1(Z)}{1 + k_1 k_2 I_1(Z) Z^{-1}} = \frac{k_1 Z}{Z - (1 - k_1 k_2)} \tag{7}$$

It can be seen from the Eq. (7) that system is equivalent to an integrator in the main circuit and a unit delay in the feedback circuit. The zero-pole of the transfer function is shown in the Fig. 5 which has a zero point (0, 0) and a pole point $(1 - k_1 k_2, 0)$. For a causal system, the condition to keep stable is that all poles are in the unit circle which means $|1 - k_1 k_2| < 1$ and the total gain is in the range of (0, 2). This is an important principle of our research. Because the hardware correlation parameters are constants for $k_1 k_2$, it is necessary to add a feedback parameter k_f in the feedback circuit to ensure stability.

3. Simulation and experiment

As k_1 and k_2 are constants, we assume $k_1 = k_2 = 1$ to simplify the analysis so the value range of k_f is (0, 2). We establish a simulation model of closed-loop FOG by Simulink as shown in Fig. 6 for different input signals and k_f. The "modu & demodu" module is a modulation and demodulation module with sinusoidal function built-in, which represents the light intensity difference shown in Eq. (5). The k_f is in the range of 0.1–1.9 with interval of 0.1 for simulation. Input signals are selected as random signal, unit step signal and constant frequency sinusoidal signal.

Also, an all-digital closed-loop FOG system is built for verification, and the hardware system is shown in Fig. 7.

FOG consists of light source module, optical coupler, Y waveguide, fiber coil, detector module, serial communication port, electrical circuit and signal processing module. The light source module includes light source and driving circuit. Detector module includes light detector, current-to-voltage conversion and preamplifier circuit. Electrical circuit and signal processing module include ADC, DAC, DSP, modulation driving circuit, etc. The fiber coil length of FOG is 2 km, the half-wave voltage is 3.5 V and the data update rate

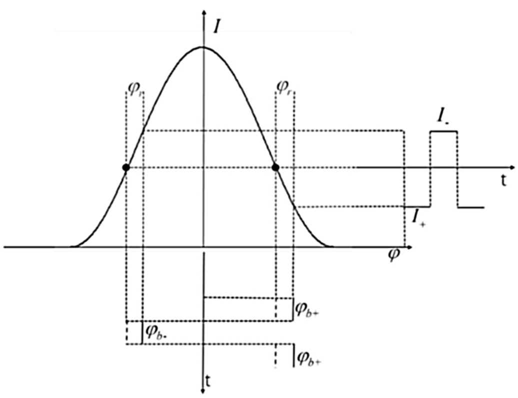


Fig. 2. Schematic diagram of phase bias and signal demodulation.

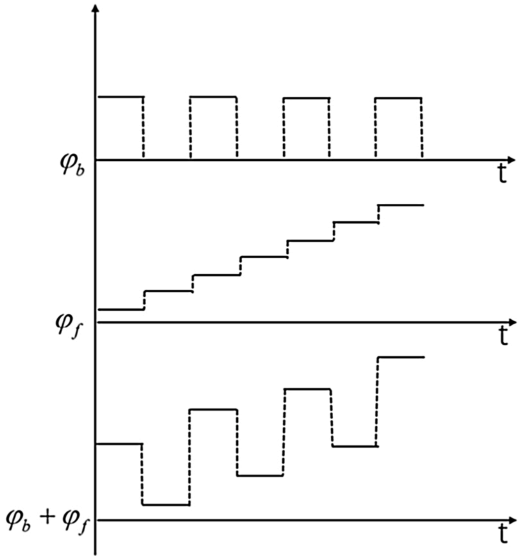


Fig. 3. The signal waveforms of modulation process.

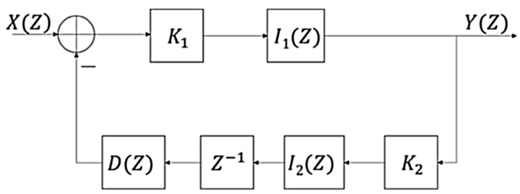


Fig. 4. Diagram of signal processing in the closed-loop control system.

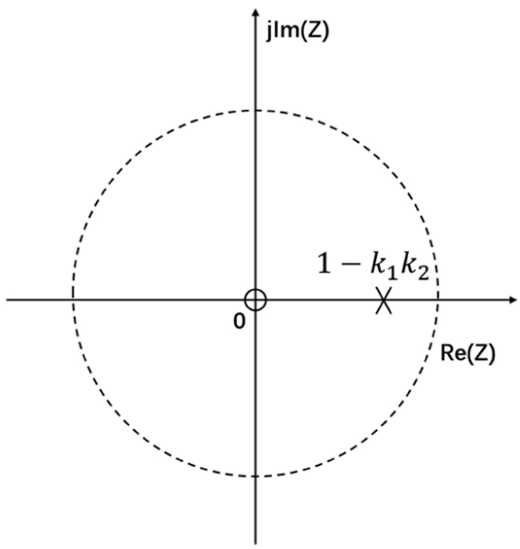


Fig. 5. The zero-pole of the transfer function for closed-loop system.

is 10 kHz.

3.1. Random signal

The random signal adopts white noise signal as shown in Fig. 8(a) with the mean value of 0, peak to peak value of 0.1 and

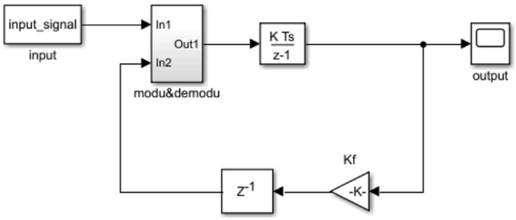


Fig. 6. Simulink model of closed-loop FOG.

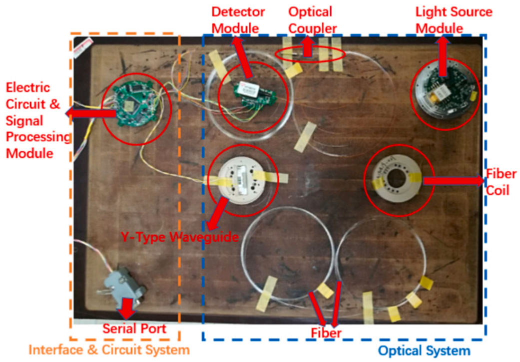


Fig. 7. The hardware system of verification experiment.

simulation time of 10000. Import it into the simulation system and outputs when the k_f is 0.1, 0.5, 1.0, 1.5 and 1.9 respectively are as shown in the Fig. 8(b).

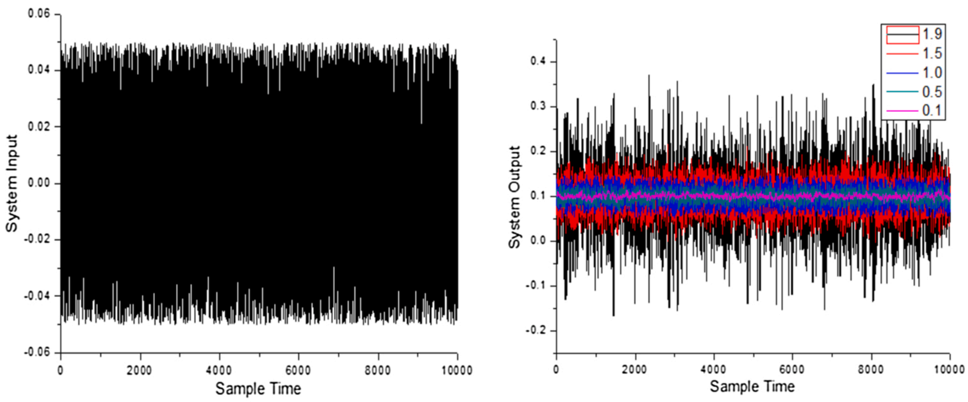
We calculate the standard deviation ratios between output signals and input signals to assess response characteristics with different k_f for random signals as shown in Fig. 9.

It can be seen from Fig. 9 that the ratios increase with higher k_f. When the k_f is less than 1.3, the ratio is less than 1 and the system has a certain inhibition effect on random noise. When the k_f is greater than 1.3, the ratio increases sharply and the system will amplify the impact of random noise. However, since the random noise added in the simulation is pseudo-random number, the output will not decrease infinitely and its trend will slow down.

When FOG works in static environment, the intensity of light propagating in optical system will appear random fluctuation to some extent which will bring random noise to the interference signal of detector and it is equivalent to random signal input. The results with partial k_f are as shown in Fig. 10.

In order to quantitatively analyze the relationship between k_f and noise, the relative standard deviation $\frac{\sigma_{K_f}}{\sigma_{0.1}}$ of the experimental and simulation data is calculated as shown in Fig. 11.

The results show that FOG noise increases monotonically with k_f and k_f is supposed to be smaller in static applications. However,



(a) **(b)**

Fig. 8. (a) Input white noise, (b) output signal with corresponding k_f.

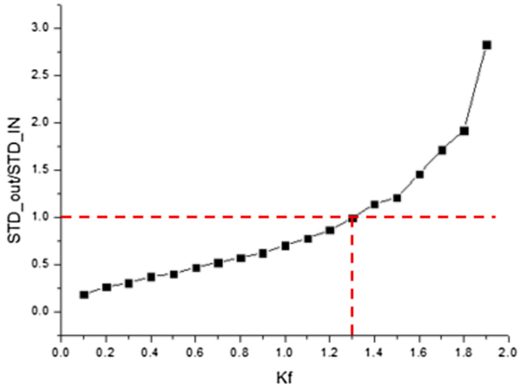


Fig. 9. Standard deviation ratios of output signals to input signals.

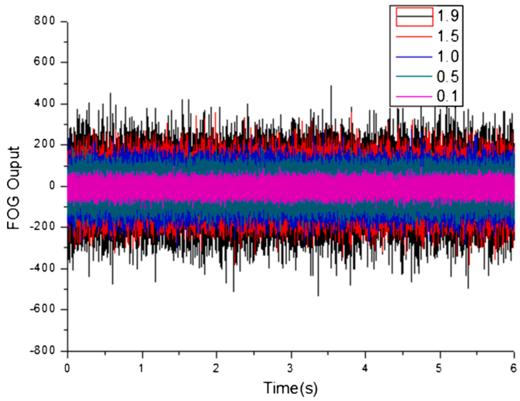


Fig. 10. Output signal of FOG for random signal input with different k_f.

the variation trend and magnitude of the experimental noise are slightly different with that of the simulation. The reasons are as follows: first of all, there is only input signal noise in the simulation which corresponds to the optical noise in FOG, while the electronic circuit noise is also introduced in experiment. Second, the input signal of simulation is pure random noise but there is also short-term instability due to the fluctuation of light source, bias error and environmental factors in experiments. Moreover, the corresponding data output frequency is consistent with the modulation frequency under simulation condition, but it is difficult to achieve the data update rate of the modulation frequency level due to the baud rate limitation of the serial communication in experiments.

3.2. Unit step signal

The unit step signal as shown in Fig. 12(a) has a simulation time of 10,000 and changes from 0 to 1 when simulation time is 5000.

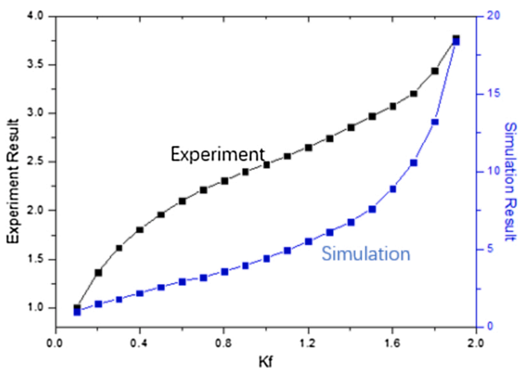


Fig. 11. Comparison of experimental and simulation results with random noise input.

The corresponding outputs with partial k_f are as shown in Fig. 12(b).

It can be seen that for unit step signal, when the k_f is small, the system will converge slowly, that is, the response will have a large delay. When the k_f is large, the system response will be faster with an overshoot phenomenon and the system output will converge after damped oscillation near the target value. In order to quantitatively evaluate the response characteristics, the convergence time is defined as the time when relative error between the system output x_O and the target value x_T is less than 0.03:

$$\frac{|x_O - x_T|}{x_T} < 0.03 \tag{8}$$

The analysis results of simulation are as shown in Fig. 13.

Whether the k_f is too small or too large, the convergence time is always long. Therefore, for unit step signal, the k_f should be selected as a moderate value.

Because the experimental system is not integrated, dynamic experiment could not be carried out. The step signal is loaded in the feedback channel as the form of software simulation. The input loading form and step signal are shown in Fig. 14.

As the experimental data update rate is 10 kHz and the corresponding data interval is 0.1 ms, we select the data of a second in the mid for analysis and the number of data points is 10000. The system outputs with different k_f are shown in the Fig. 15.

The output digital quantity of FOG after convergence is about 11000. It is because the FOG digital quantity is accumulated and shifted during demodulation and output, so there is a conversion coefficient between the output and the input calculated as 0.344. For the step signal, when the k_f is small, the system will converge slowly as simulation. When the k_f is large, the overshoot phenomenon is submerged to a certain extent because of the existence of random noise, but it is still visible near the step point and the output converges to the target value finally. The evaluation during simulation is slightly adjusted when assessing the response characteristic quantitatively. The algorithm is as follows: take the data of 10 s after the step, calculate the mean value and extreme value with setting them as x_{ave}, $x_{\max}$ and $x_{\min}$. Perform a reverse order search from the 1000th point after the step to the step point, the time difference between the first point and the step point that meets the following conditions is convergence time:

$$\begin{cases} x_O < x_{min} - 0.03 * (x_{max} - x_{min}) \\ or \\ x_O > x_{max} + 0.03 * (x_{max} - x_{min}) \end{cases} \tag{9}$$

According to the above algorithm, the simulation and experimental results are analyzed as shown in the Fig. 16.

For the application with step signal or other high response requirements, the k_f should be moderate such as (0.6, 1.4).

3.3. Constant frequency sinusoidal signal

According to the theory of Fourier transform, any time domain signal can be expressed as superposition of sinusoidal signals with different frequencies, it is of practical significance to analyze the response characteristics for sinusoidal signal. Furthermore, as the detection signal usually contains errors of different frequencies, it can also be used for error compensation such as vibration especially in closed-loop scheme [19]. The sinusoidal signal with amplitude of 1 and bias of 1 is used. The simulation time is 10,000 and the expression of input signal is:

$$y = 1 + \sin(2\pi ft) \tag{10}$$

Here f is its frequency. The modulation frequency of FOG is related to the length of fiber coil and modulation mode. For a FOG with the fiber coil length of about 1 km, the modulation frequency is generally dozens or hundreds of kHz. The frequency of the signal to be measured in the application scenario and the actual bandwidth demand is far less than this modulation frequency. To facilitate analysis, the input signal frequencies are normalized to modulation frequency which are set as 0.0001, 0.001, 0.01, 0.05, 0.1, 0.15 and 0.2. When the normalized input signal frequencies are 0.001 and 0.1 the outputs are as shown in the Fig. 17.

When the frequency of input is low, all k_f can achieve good tracking effect but small k_f has a little delay. When the frequency is high, the output and the input signal have good consistency only when k_f is large, that is, the signal waveform distortion and delay effect are small.

The simulation results of all input signal frequencies and k_f are analyzed in frequency domain and compared with input. The frequencies of their results are always consistent with input signals, but the amplitudes and phases of different k_f are very different. The normalized amplitude ratios and the phase differences of output and input are shown in the Fig. 18.

It can be seen that there is serious signal distortion which will greatly reduce the reliability and accuracy of measurement when the k_f is small for high dynamic application environment. Only when the k_f is large, their amplitudes and phases are in good consistency with the input signal, that is, a large k_f can ensure the authenticity and integrity of the signal in the measurement bandwidth of FOG.

The sinusoidal signal loaded in the feedback channel is also in the form of software simulation which is sampled as 20 points of a period. When the signal is output, the sampling values are obtained by looking up the table based on time series. The sinusoidal signal is shown in Fig. 19.

The modulation frequency of FOG is about 25 kHz and the data update rate is 10 kHz. According to the hardware characteristics and program development requirements of the lower computer, the sinusoidal signal frequencies are set as 2.46 Hz, 24.64 Hz, 246.44 Hz and 2217.93 Hz, which means the actual normalized frequencies are 0.0001, 0.0010, 0.0099 and 0.0887 respectively. When the

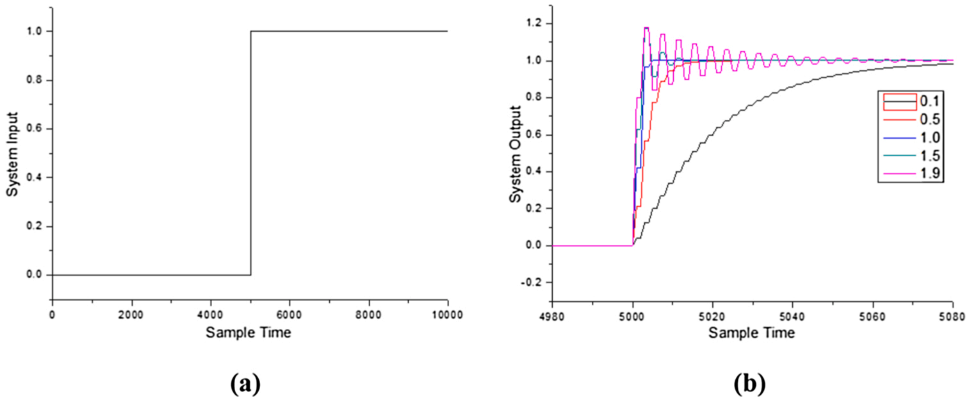


Fig. 12. (a) Input unit step signal, (b) output signal with corresponding k_f.

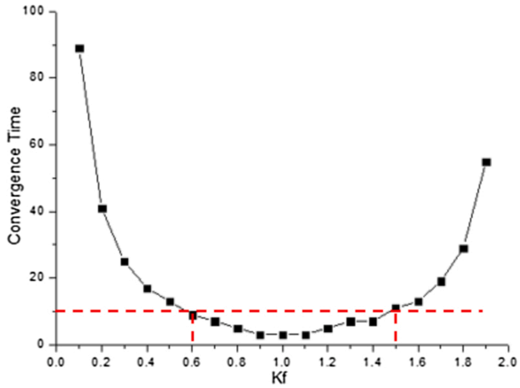


Fig. 13. Convergence time of output signal with different k_f.

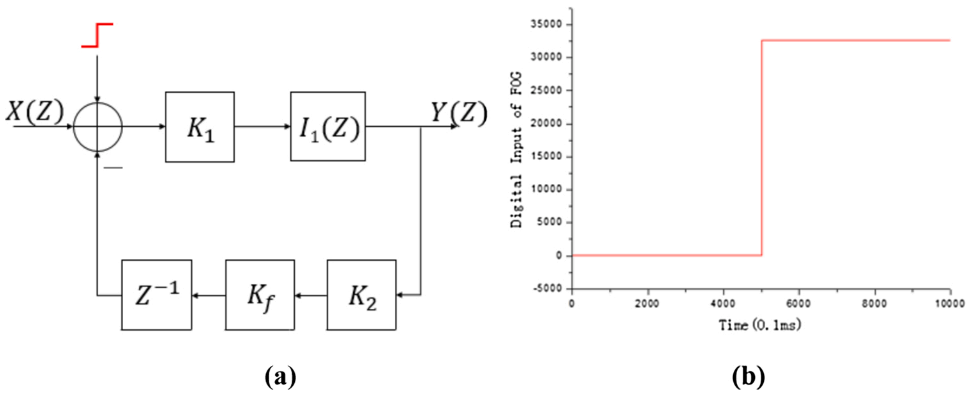


Fig. 14. (a) Loading form of input signal, (b) step signal in experiments.

frequencies are 24.64hz and 2217.93hz, the output signals for partial k_f are shown in the Fig. 20.

When the low frequency signal is input, the influence of random noise is not considered. For different k_f, the output signal is basically the same as the input, but the short-term noise will be large when the k_f is higher. When the high frequency signal is input, the FOG still has good tracking ability as the k_f is large, but the distortion and delay are serious when the k_f is small. The experimental results are analyzed in frequency domain as mentioned above. The results are shown in Fig. 21.

Limited by the data update rate of FOG, the frequency domain analysis will be greatly affected when the frequency is high. If the system output simultaneously meets the requirements of normalized amplitude is greater than 0.97 and phase difference is greater than -0.1π, the signal distortion can be ignored. The k_f satisfying the above two conditions under different input signal frequencies are analyzed. The results are as shown in Fig. 22 and the shaded part indicates the range of the k_f that output signal is basically undistorted

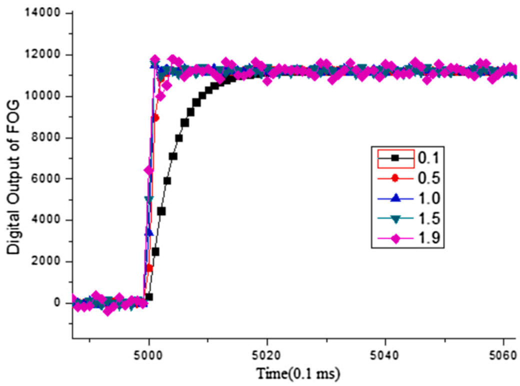


Fig. 15. Output signal for step signal with different k_f.

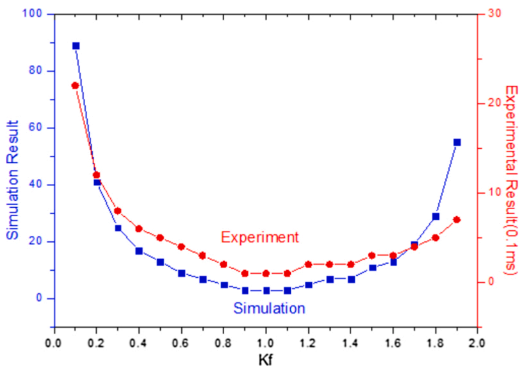


Fig. 16. Comparison of experimental and simulation results with step signal input.

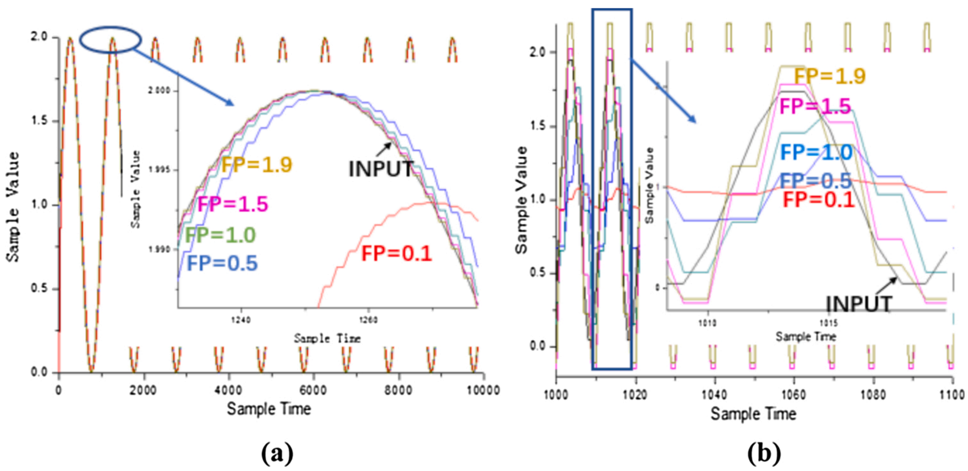


Fig. 17. Partial simulation results with sinusoidal signal: (a) $f = 0.001$, (b) $f = 0.1$.

at different frequencies.

It can be seen from Fig. 22 that different k_f should be selected to meet the requirements for specific dynamic applications. For example, when the FOG is used in navigation and other low-bandwidth measurement fields, the k_f should be relatively small such as (0.6, 1.4) which can not only ensure the response performance of low frequency signal but also suppress high frequency noise. When FOG is used in high-bandwidth fields like satellite flutter measurement, the k_f should be larger such as (1.5, 1.9) to ensure the signal integrity.

It should be noted that the maximum data refresh rate is only 10 kHz due to the hardware limitation. For high frequency input

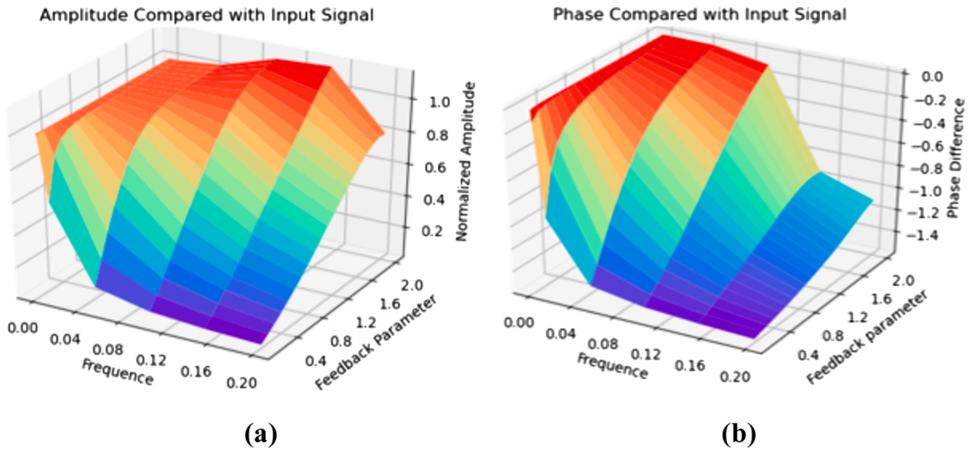


Fig. 18. Simulation statistical results: (a) normalized amplitude, (b) phase difference.

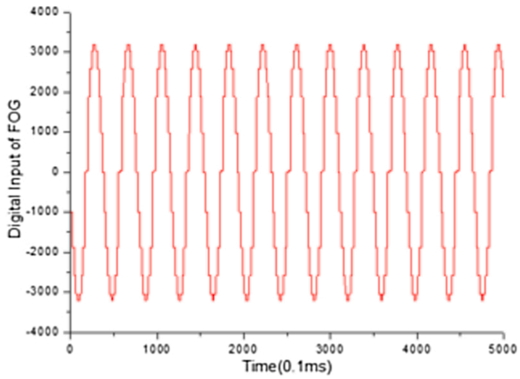


Fig. 19. Sinusoidal signal in experiments.

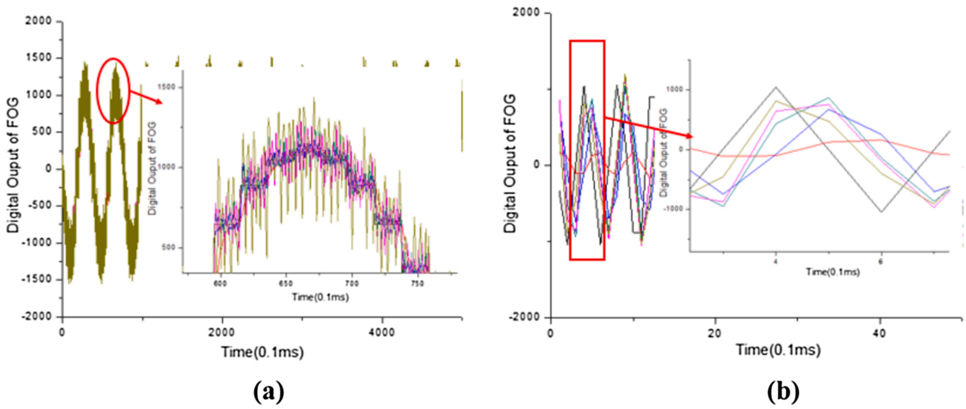


Fig. 20. Output signals for sinusoidal signal with different k_f: (a) $f = 24.64$Hz, (b) $f = 2217.93$Hz.

signals, the number of periodic sampling points is not enough which will affect the integrity of the signal and also limit the development of higher frequency experiments designed in the simulation. However, the experimental results are in good agreement with the simulation analysis and the factors limited by the experimental conditions will not have a substantial impact on the conclusion of the sinusoidal signal.

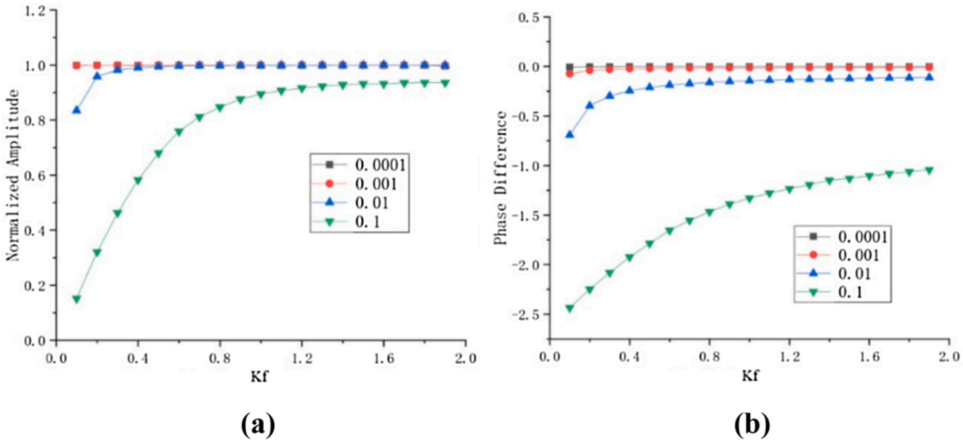


Fig. 21. Statistical experimental results after processing: (a) normalized amplitude, (b) phase difference.

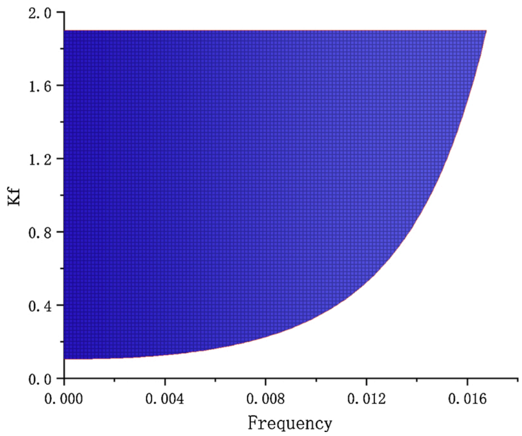


Fig. 22. The range of k_f for sinusoidal signals with limited distortion.

4. Conclusion

This paper establishes a concise feedback control model of all-digital closed-loop FOG and obtains the range of feedback parameter k_f. The output effects of three typical signals with different k_f are simulated by Simulink and assessed. Although there are some unavoidable disturbances, relevant experiments show that the model and simulation have high credibility. It could be proved that setting the appropriate k_f will greatly improve the performance of FOG.

It can be seen from Section 3 that the k_f is supposed to be smaller in static applications and as moderate as (0.6, 1.4) for high response requirements. What is more, as discussed in the sinusoidal input signal, when FOG is used in low-bandwidth measurement fields, the k_f should be relatively small to ensure the response performance of low frequency signal and suppress high frequency noise. When FOG is used in high-bandwidth fields, the k_f should be large such as (1.5, 1.9) to ensure signal integrity. Because any signal can be expressed as superposition of sinusoidal signals with different frequencies and some errors can also be compensated by this way, this conclusion and related experimental data are of great significance. Researchers can set k_f according to the specific hardware conditions.

At present, improving the performance of closed-loop FOG has attracted extensive attention and we have been devoted to optimizing the essential closed-loop feedback parameter for different input signals by carrying out groups of simulations and experiments to fill the gap in this field. As more and more sensors designed in various fields adopt closed-loop scheme, it will help the researchers to adjust the feedback control chain in the sensor field, especially in closed-loop FOG.

Author agreement

All authors have read and approved to submit it to your journal. There is no conflict of interest of any authors in relation to the submission. The paper has not been submitted elsewhere for consideration of publication.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work.

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